

Luke McEvoy

Lukemcevoy7@hotmail.com | New York, NY | (631) 478-9254
<https://www.linkedin.com/in/luke-mcevoy> | <https://github.com/luke-mcevoy>

EDUCATION

Stevens Institute of Technology | Hoboken, NJ

Ph.D., Physics,	GPA: 3.65 / 4.0	Expected May 2026
M.S., Machine Learning,	GPA: 3.83 / 4.0	May 2023
B.S., Computer Science,	GPA: 3.52 / 4.0	May 2022

Awards: Eagle Scout, Presidential Scholarship, Dean's List, Best Student Paper Presentation at SPIE Photonics Europe
Coursework: Algorithms, Data Structures, Project Management, Intermediate Statistics, Probability & Statistics, Database Management, Systems Programming, Linear Algebra, Agile Development, Operating Systems, Calculus (1,2,3), Quantum Mechanics, Statistical Mechanics, Mechanics, Machine Learning, Artificial Intelligence, Deep Learning, Quantum Computing

SKILLS

Languages: Python, C++, Golang, C, JavaScript, TypeScript, SQL, Java, R, MATLAB

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, OpenCV, React.js, Next.js, Node.js

Cloud & Infrastructure: AWS, GCP, Kubernetes, Docker, Argo CI/CD, Kafka, Redis, Linux, VxWorks, REST APIs

Optics & Photonics: Single-Photon LiDAR (SPAD/CMOS detectors, ToF histograms), ultrafast lasers (mode-lock laser), DWDM filters, MEMS beam steering, Nonlinear optics, FPGA, quantum frequency conversion, single-photon vibrometry

WORK EXPERIENCE

Software Engineer Intern, [Tesla](#) | Palo Alto, CA

6/25–9/25

- Made Tesla Autopilot's AI map generation pipeline **350x faster** by parallelizing satellite/GPS/label data ingestion across Kubernetes pods, cutting preprocessing of 400k images from **56 hours to 10 minutes**.
- Deployed robotaxi user behavior auditing system that evaluates rider activity and assigns reliability scores to users using Golang, Ruby, TypeScript, React.js, Next.js, Kafka, AWS, Kubernetes, and Docker.
- Engineered a maintenance geofencing system that automatically opens garage doors when Robotaxis enter designated zones, leveraging real-time telemetry, asynchronous message handling, and geospatial logic.
- 2nd highest contributor (lines of code and commits) on a 10-engineer team of senior and staff software engineers rewriting Tesla's authentication codebase using TypeScript, Next.js, Golang, React.js, and Docker.
- Optimized Argo CI/CD via parallelization, cutting test time by 50% and saving ~60 engineering hours per day.

Machine Learning Engineer, [Quantum Computing Inc.](#) | Hoboken, NJ

5/23–5/24

- Designed ML/AI pipelines for Quantum Single-Photon LiDAR, achieving 99% material recognition accuracy through SPAD-based time-of-flight histograms, maintaining **89% accuracy through 15.2 OD obscurants**.
- Developed a nonlinear optical system for a Single-Photon Vibrometer, integrating 1554-1585 nm picosecond pulses, DWDM filtering, MEMS beam steering, and SPAD detectors for applications in landmine detection, long-range voice sensing, structural health monitoring, elastography, and surveillance.
- Built nonlinear optics and ML models for surface roughness metrology using single-photon LiDAR speckle statistics, measuring roughness from **1.2 μm to 102 μm at range**.
- Developed an emotion classification model using an FPGA-based reservoir computer which outperformed the results of Million Ways, an emotional intelligence AI startup QCI debated acquiring. QCI passed on acquisition.

Research/Teaching Assistant, [Stevens Institute of Technology](#) | Hoboken, NJ

1/23–Present

- Fully funded PhD program by a \$15,000,000 Department of Defense grant.
- Taught 3 classes of Mechanics (PEP 111) containing 80+ students
- Invented the Physics-Informed Masked Autoencoder (PI-MAE), enabled single-photon LiDAR reconstruction with as few as **9 photons/pixel at 90% masking**, outperforming Navier-Stokes and Fast Marching Method.
- Developed VideoPI-MAE, model for 3D single-photon LiDAR video, achieving **50x faster imaging** under 98% masking with MEMS-driven sparse sampling and spatiotemporal transformers
- Built and operated full Single-photon LiDAR optical stack: 6 ps mode-locked lasers at 1554-1556 nm, DWDM filters, MEMS beam steering, SPAD detector, Periodically Poled Lithium Niobate waveguide, FPGA controller.
- Coded / applied PointNet++, CNNs, SAM, Transformers, Autoencoders for sparse-photon 3D reconstruction.

Software Engineer Machine Learning Intern, [Lenovo](#) | Raleigh, NC

1/22–6/22

- Designed and developed full stack client acquisition application that discovers and visualizes financially lucrative Uniform Commercial Code (UCC) and SEC filings for the Global Lenovo TruScale Offerings sales representatives to scalp competitors' expiring leases.

U.S. Citizen

- Created algorithm which scrapes UCC and SEC filings from US federal websites to feed application data.
- Strategic Initiative Lab Winter Analyst, [RBC Capital Markets](#)** | Jersey City, NJ 1/22-1/22
- Developed and deployed peer to peer hybrid cloud infrastructure that is robust to cloud outages using Python, JavaScript, Django, React, Node.js, Google Cloud, AWS, Kubernetes, and Docker.
 - Increased cloud coverage by 3000% compared to previous system architecture.
- Software Engineer Intern, [ASML](#)** | San Diego, CA 6/21-8/21
- Implemented a multithreaded streaming interface in ASML's \$160,000,000 Extreme UV machine, making the signal writing process **1,667% more efficient**. Engineered with C, C++, Python, GTest, Linux, VxWorks.
 - Designed and implemented extensions to a hardware simulator for the embedded systems of the Extreme UV lithography light source using MATLAB, C, C++, Python, and company specific languages.
 - These hardware simulator extensions save Software Engineers an estimated **240 hours per year**.
 - Used C, C++, Python, and UML skills to analyze complex issues, audit software architecture, determine the best testing methodologies, and contribute to all aspects of the software development build, test and release life cycle.
- Chief Technology Officer, [CastlePoint Analytics](#)** | Hoboken, NJ 8/20-6/21
- Built the MVP that helped secure first round VC funding from investors, including Fox Sports NFL Analysts.
 - Developed Machine Learning model that predicted **MMA fights 72%**, earning 160 paying customers for metrics.
 - Product Owner and Lead Developer of Web Application to distribute our analytics. Lead 6 other Engineers.
 - Exited CastlePoint Analytics after it was acquired in May 2021 by a Banker from Jefferies Group.
- Social Media Market Intelligence Extern, [Facebook](#)** | Remote 2/21-3/21
- Collaborated with Facebook Disruptors team to grow Direct-to-Consumer eCommerce companies.
 - Researched companies' use of competitive social media, competitive advertising tools, and overall market trends.
 - Clients included HelloFresh, Paramount, Fox, Peloton, Casper, Bombas, Roman, Noom, HomeChef.
- Android Engineer Intern, [WW](#)** (formerly Weight Watchers) | New York City, NY 6/20-7/20
- Built out two Machine Learning features to the production application with **2,000,000+ active users** using Kotlin, Git, Android, Jira, Confluence, Retrofit, RxJava, Moshi, CameraX, and Google ML Kits.
 - Published production application features were Barcode Scanner and OCR NutritionLabel Scanner.

PUBLICATIONS

L. McEvoy, D. Tafone, Y. Meng Sua, and Y. Huang, "Video Physics-Informed Masked Autoencoder for High-Speed Photon Imaging of Dynamic 3D scenes" CLEO Applications and Technology (2025) <https://doi.org/10.1364/CLEO>

L. McEvoy, D. Tafone, Y. Meng Sua, and Y. Huang, "Physics-Informed Masked Autoencoder for Active Sparse Imaging", Scientific Reports (2024) <https://doi.org/10.1038/s41598-024-71095-x>

[Best Student Paper Presentation Award] **L. McEvoy**, D. Tafone, Y. Meng Sua, and Y. Huang, "Inpainting sparse scenes through physics aware transformers for single-photon LiDAR", in SPIE Photonics Europe (2024) <https://doi.org/10.111>

L. McEvoy, D. Tafone, Y. Meng Sua, and Y. Huang, "Sparse Single Photon Reconstruction through Masked Autoencoder", in Frontiers in Optics + Laser Science, paper JW4A.86 (2023) <https://doi.org/10.1364/FIO.2023.JW4A.86>

D. Tafone, **L. McEvoy**, Y. Meng Sua, P. Rehai, and Y. Huang, "Surface material recognition through machine learning using time of flight LiDAR," Opt. Continuum 2, 1813-1824 (2023) <https://doi.org/10.1364/OPTCON.492258>

D. Tafone, **L. McEvoy**, Y. Meng Sua, P. Rehai, and Y. Huang, "Material recognition using time of flight Lidar surface analysis" Proc. SPIE Quantum Sensing, Imaging, and Precision Metrology, (2023) <https://doi.org/10.1117/12.2652945>

PERSONAL PROJECTS

- Podcast Founder & Host, Collision Theory** 3/20-1/24
- Created a podcast where I interview industry professionals from Facebook, Twitter, Google, BlackRock, Prudential and more regarding their professional development and lessons. 500 listeners from 6 continents to date.
- Solo Backpacking** 5/19-Present
- Have been to 32 countries across 4 continents by budget traveling and my disciplined saving habits.